CHECKLIST OF COMPLIANCE WITH THE SYLLABUS PARTS AND FORMAT

Course Code and Title: **S-CSPE003LC-Parallel and Distributed Computing**

This undertaking has a two-fold objective. First, it allows the syllabus designers to review their output systematically. Based on their honest evaluation, they could make the necessary revision. Second, it expedites the checking to be done by the Dept. Syllabus Committee Reviewer, Department Chair, the College Dean, and the Institutional Syllabus Reviewers.

DIRECTIONS: Review the latest version of the syllabus that you submitted to the Department Chair for his/her scrutiny. As you do your reflective review, As you do your reflective review, assess your compliance with the syllabus parts and format by answering each of the items that follow. Shade the corresponding box (complied with, not complied with) under the compliance column.

Format, parts, and description	Compliance	
	Complied With	Not Complied With
Format		
1. The 2023-2024 syllabus format is used.	/	
2. The font used in the syllabus preparation is Times New Roman (11).	/	
3. DLSUD (being first) and college logos are beside each other at the top (center).	/	
4. Preliminaries include all the contact details of the professor, such as the day and time.	/	
5. Each WEEK has a corresponding date.	/	
Content		
6. The Learning Outcomes have more HOTS than LOTS.	/	
7. CLOs and TLOs are identified for each module/lesson/topic.	/	
8. Gospel reading and Reflection and reflection precede the presentation of each topic under the teaching-learning activities.	/	
9. The journal articles/references are within the last five years.	/	

10. Some materials are beyond the five-year rule because these are considered cannon and seminal studies.	/	
11. The course policies and requirements section of the syllabus is a clear provision for addressing the concerns and grievances of students.	/	
Assessments		
12. At least three and a maximum of five Enabling Assessments per term are reflected in the syllabus.	/	
13. Formative Assessments are not reflected anymore in the syllabus.	/	
14. Enabling Assessments are briefly described and aligned with the learning outcomes and teaching-learning activities.	/	
15. The syllabus presents one-module(lesson)-one-assessment correspondence.	/	
16. Only one Enabling Assessment is reflected in a given week	/	
17. The Enabling Assessments are aligned with the identified CLOs and TLOs.	/	
18. Enabling Assessments, Class Participation, and Summative Assessments constitute the breakdown of grades per term.	/	
19. Thirty percent (30%) is allotted for Summative Assessments per term.	/	
20. Fifteen percent 20%) is allotted for Class Participation per term.	/	
21. The percentage that is allocated for Enabling Assessments fairly corresponds to their quantity, depth, breadth, and level of difficulty compared to the other components of the Midterm Grade and Final Grade.	/	
22. Summative Assessments are aligned with the identified CLOs, TLOs and teaching-learning activities.	/	
23. Enabling Assessments are aligned with the identified CLOs and TLOs and teaching-learning activities.	/	
24. The Midterm Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire MIDTERM PERIOD	/	
25. The Final Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire FINAL PERIOD.	/	
References and Curriculum Map		
26. The References must be within the last 5 years except for canons and seminal studies.	/	
27. It must be a combination of books, eBooks, journal articles, videos, etc.	/	

28. It should follow the APA 7th edition for CLAC, CBAA, CTHM, CCJE, CEAT, COED, and CSE for CSCS.	/	
29. The syllabus is submitted with an attached mapping table (CLOs with PLOs and ILOs)	/	

NOTE: Enabling Assessments include Discussion Type (cf. SB), Forum, Analysis Paper, Reports, etc.

Self-reviewed:



Jennylinde R. Manaois

Faculty

Checked, verified, and endorsed:



Rolando B. Barrameda

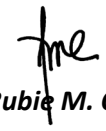
Dept. Curriculum Committee Reviewer

Checked, verified, and approved:



Josephine T. Eduardo

Department Chair



Dr. Rubie M. Causaren

College Dean

DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF SCIENCE AND COMPUTER STUDIES
COMPUTER STUDIES DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	S-CSPE003LC
COURSE TITLE	**:**	Parallel and Distributed Computing
COURSE TYPE	**:**	Laboratory
COURSE CREDIT	**:**	1 Unit (3 hours)
MODALITY	**:**	Onsite learning model
PRE-REQUISITES	**:**	Should be 3rd year level
CO-REQUISITES	**:**	None
PROFESSOR	**:**	Jennylinde R. Manaois
CONSULTATION HOURS	**:**	M / 2:00 -4:00, T / 1:00 -4:00, TH 9:00 – 12:00, 2:00 – 4:00 F / 1:00 – 2:00 jrmanaois@dlsud.edu.ph

COURSE DESCRIPTION:

Covers widely used parallel and distributed computing methods, including threaded applications, GPU parallel programming, and datacenter-scale distributed methods such as MapReduce and distributed graph algorithms. Different types of algorithms are discussed which work well with these techniques, and have the opportunity to implement some of these algorithms. Hardware architectures which have been developed along with these computing methods are be included.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. know that parallel computation is a natural means to model the real world, and have a basic understanding of programming and coordinating multiple processes to execute simultaneously to accomplish some task;

CLO2. apply the demonstrated techniques to new computational problems, and generally be able to maintain intelligent conversation on parallel computing;

CLO3. use common thread libraries for new problems and also have a working knowledge of some basic concepts of parallel programming;

CLO4. provide a framework for introducing PDC topics while teaching digital logic;

CLO5. assess the concept of collective communication and the need for unique identifiers for each computational process; and

CLO6. demonstrate 21st century skills in all learning activities.

LEARNING PLAN :

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Know that parallel computation is a natural means to model the real world, and have a basic understanding of programming and coordinating multiple processes to execute simultaneously to accomplish some task.	• TLO1. Understand the basic concepts of parallel computation.
CLO3. Use common thread libraries for new problems and also have a working knowledge of some basic concepts of parallel programming.	• TLO5. Write small parallel programs in terms of explicit threads that communicate via shared memory and synchronize via fork and join;
CLO4. Provide a framework for introducing PDC topics while teaching digital logic.	• TLO2. Recognize basic interconnection structures like tree, bus, mesh, and torus. • TLO3. Identify and apply parallel and sequential structures in hardware
CLO5. Assess the concept of collective communication and the need for unique identifiers for each computational process.	• TLO4. Assess the existence of message passing paradigms for cluster computing.
CLO6. Demonstrate 21st century skills in all learning activities.	• Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

Module No and Learning Outcomes	**Date**	**Teaching-Learning Activities / Assessment Strategies**	**Technology Enabler**	**Onsite**	**Allotted Hours**
Module 1 CLO 6	Week 1 (Aug.14-19)	• Gospel reading • Syllabus uploaded • Presentation of course introduction	Schoolbook, compiler	/	3
Module 2 TLO 1,2,3 CLO1,2	Week 2 (Aug. 21-Aug. 26)	• Gospel reading • Introducing parallel and distributed concepts in digital logic	Schoolbook, compiler	/	3

Module 2 TLO 1,2,3 CLO1,2	Week 3 (Aug. 28-Sept. 2)	• Gospel reading • Introducing parallel and distributed computing concepts in digital logic	Schoolbook, compiler	/	3
Module 3 TLO 4 CLO 5	Week 4 (Sept. 4-9)	• Gospel reading • Networks and parallelism with a data-structures focus	Schoolbook, compiler	/	3
Module 2,3	Week 5 (Sept. 11-16))	• Long Quiz 1: Module 2,3	Schoolbook, compiler	/	3
Module 4 TLO 5 CLO 3	Week 6 (Sept. 18-23)	• Gospel reading • Fork-join parallelism with a data-structures focus	Schoolbook, compiler	/	3
Module 4 TLO 5 CLO 3	Week 7 (Sept. 25-30)	• Gospel reading • Fork-join parallelism with a data-structures focus	Schoolbook, compiler	/	3
Module 4 TLO 5 CLO 3	Week 8 (Oct. 2-7)	• Gospel reading • Fork-join parallelism with a data-structures focus	Schoolbook, compiler	/	3
Module 2,3,4 TLO 1,2,3,4,5 CLO 1,2,3	Week 9 (Oct. 17-22)	• Midterm Examination	Schoolbook, compiler	/	3
			TOTAL	***27***	***27.0***

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Know that parallel computation is a natural means to model the real world, and have a basic understanding of programming and coordinating multiple processes to execute simultaneously to accomplish some task.	TLO10. Understand basic concepts in parallel programming: data/vector/loop-based parallelism, as well as the basics of distributed memory programming and task scheduling.
CLO2. Apply the demonstrated techniques to new computational problems, and generally be able to maintain intelligent conversation on parallel computing.	• TLO6. Use locks to implement critical sections correctly. • TLO7. Follow basic guidelines for easier concurrent programming, such as avoiding mutable thread-shared state where possible and following consistent protocols for what locks protect what data. • TLO8. Break down a complex computation (quicksort) into individual parallel operations.
CLO5. Assess the concept of collective communication and the need for unique identifiers for each computational process.	• TLO9. Assess costs of parallel operations in a shared-memory environment

CLO6. Demonstrate 21st century skills in all learning activities.	• Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

Module No and Learning Outcomes	**Date**	**Teaching-Learning Activities / Assessment Strategy**	**Technology Enabler**	**Onsite**	**Allotted Hours**
Module 2,3,4 TLO 1,2,3,4,5 CLO 1,2,3	Week 10 (Oct. 16-21)	• Completion of missed assessments	Schoolbook, compiler	/	3
	Week 11 (Oct. 23-28)	Self-care week			3
Module 4 TLO 6,7 CLO 2	Week 12 (Oct. 30 – Nov. 4)	• Gospel reading • Shared-memory concurrency control w/ a data-structures focus	Schoolbook, compiler	/	3
Module 4 TLO 6,7 CLO 2	Week 13 (Nov. 6 – 11)	• Gospel reading • Shared-memory concurrency control w/ a data-structures focus	Schoolbook, compiler	/	3
Module 5 TLO 8,9 CLO 2,5	Week 14 (Nov. 13 – 18)	• Gospel reading • Parallel computing in a Python-based Computer Science course	Schoolbook, compiler	/	3
Module 4,5 TLO 6,7,8,9 CLO 2,5	Week 15 (Nov. 20 – 25)	• Long Quiz : Module 7, 8, 9,10	Schoolbook, compiler	/	3
Module 6 TLO 10 CLO 1	Week 16 (Nov.27 – Dec. 2)	• Gospel reading • Parallel programming illustrated through Conway's Game of Life	Schoolbook, compiler	/	3
Module 6 TLO 10 CLO 1	Week 17 (Dec. 4-9)	• Gospel reading Parallel programming illustrated through Conway's Game of Life	Schoolbook, compiler	/	3

Module 4,5,6 TLO 6,7,8,9,10 CLO 1,2,5	Week 18 (Jan. 9 – 14)	• Review / Consultation of Module 5,6,7	Schoolbook, compiler	/	3
Module 4,5,6	Week 18 (Dec 11-16)	• Final Examination	Schoolbook, compiler		3
			TOTAL	***27***	***27***

GRADING SYSTEM :

Midterm

Enabling Assessments - 50%
Attendance/Class Participation - 20%
Summative Assessment - 30%
100%

Finals

Enabling Assessments - 50%
Attendance/Class Participation - 20%
Summative Assessments - 30%
100%

Overall Final Grade = $\frac{\text{Midterm + Final}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Onsite Learning Mode. All activities will be done in the laboratory classroom and field.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar.

3. Laboratory Policies
   a. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
   b. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones also cannot be used as calculator during examination.
   d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

4. Communication and Feedback
   a. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for

asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:

i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
iii. Email – you may email your professor. (Professor's email will be given during the course orientation)

b. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

5. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

6. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is strictly prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

7. Other Policies
   a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.

REFERENCES :

Online References

E-provider	References
https://www.khanacademy.org/computing/ap-computer-science-principles/algorithms-101/x2d2f703b37b450a3:parallel-and-distributed-computing/a/parallel-computing	Parallel Computing
https://www.khanacademy.org/computing/ap-computer-science-principles/algorithms-101/x2d2f703b37b450a3:parallel-and-distributed-computing/a/distributed-computing	Distributed Computing

https://blog.purestorage.com/purely-informational/parallel-vs-distributed-computing-an-overview/	Parallel vs. Distributed Computing: An Overview
https://www.youtube.com/watch?v=q7sgzDH1cR8	Parallel Computing Explained In 3 Minutes
https://www.youtube.com/watch?v=rYK-kTBUrK4	Distributed Systems

Onsite References:

Provider	References
AEA Lib	Sushil K. Prasad, Anshul Gupta, Arnold L. Rosenberg, Alan Sussman and Charles C. Weems. (2015). Topics in parallel and distributed computing. ScienceDirect Electronic books
AEA Lib	**Gerassimos Barlas**. (2015). **Multicore and GPU programming : an integrated approach**. Amsterdam : Morgan Kaufmann.

Prepared: **Computer Studies Department**
1st Semester 2023-2024



Approved: **Josephine T. Eduardo**
Chair, CSD



Endorsed: **Rubie M. Causaren, PhD**
Dean, CSCS

DE LA SALLE UNIVERSITY - DASMARIÑAS
COLLEGE OF SCIENCE AND COMPUTER STUDIES
COMPUTER SCIENCE DEPARTMENT

PROGRAM LEARNING OUTCOME - COURSE LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

COURSE CODE	**:**	S-CSPE003LC
COURSE TITLE	**:**	Parallel and Distributed Computing
COURSE TYPE	**:**	Laboratory
COURSE CREDIT	**:**	1 Unit
MODALITY	**:**	Onsite learning model
PREREQUISITE	**:**	None

Course Learning Outcome	**Program Learning Outcomes**							
	PLO 1	**PLO 2**	**PLO 3**	**PLO 4**	**PLO 5**	**PLO 6**	**PLO 7**	**PLO 8**
1. Know that parallel computation is a natural means to model the real world, and have a basic understanding of programming and coordinating multiple processes to execute simultaneously to accomplish some task.	✔	✔	✔	✔				
2. Apply the demonstrated techniques to new computational problems, and generally be able to maintain intelligent conversation on parallel computing.	✔	✔	✔					
3. Use common thread libraries for new problems and also have a working knowledge of some basic concepts of parallel programming.	✔	✔	✔	✔				
4. Provide a framework for introducing PDC topics while learning digital logic.	✔	✔	✔	✔		✔		
5. Assess the concept of collective communication and the need for unique identifiers for each computational process.	✔	✔	✔	✔	✔	✔	✔	
6. Demonstrate 21st-century skills in all learning activities.					✔	✔	✔	✔

NOTE: Provide a checkmark on the areas in which the program learning outcome (PLO) is hit by the course learning outcome (CLO)

DE LA SALLE UNIVERSITY - DASMARIÑAS
COLLEGE OF SCIENCE AND COMPUTER STUDIES
COMPUTER SCIENCE DEPARTMENT
GRADUATE ATTRIBUTES (DESCRIPTORS/INSTITUTIONAL LEARNING OUTCOMES) – PROGRAM LEARNING OUTCOME
MAPPING TABLE FOR BS COMPUTER SCIENCE

Program Learning Outcomes	Graduate Attributes/ Institutional Learning Outcomes				
	GA1	GA2	GA3	GA4	GA5
1. Apply concepts of computing and mathematics in different domains of computer science.				✔	✔
2. Identify and analyze problems and define computing requirements appropriate to their solutions.					✔
3. Design, develop, test, and evaluate various solutions using **algorithms** for complex computing problems that take full advantage of the usability of computer resources.			✔	✔	✔
4. Document computer-based systems by writing effective reports for risk-assessment impact.				✔	✔
5. Work effectively as an individual or as a member of a team during software development.			✔	✔	✔
6. Communicate effectively with the computing community and non-computer science professionals about computing activities.				✔	✔
7. Recognize the legal, social, ethical, and professional issues involved in the utilization of computer technology and address these issues.	✔	✔	✔		
8. Engage in continuing professional development through performing independent or collaborative research as computing professionals.			✔	✔	✔

NOTE: Provide a checkmark on the areas in which the Graduate Attribute (Descriptors/Institutional Learning Outcome) is hit by the program learning outcome (PLO). Kindly refer to the descriptors (institutional learning outcomes) to clearly understand what each attribute refers to or expects from its graduates.

GRADUATES ATTRIBUTES AND INSTITUTIONAL LEARNING OUTCOMES (ILOs)

Graduate Attribute (GA)	Descriptors (Institutional Learning Outcome)[1]
1. God-centered	a) Creates an environment where the experience of God is lived and shared b) Practices honesty, fairness, truth, and integrity in all aspects of life (personal and professional lives) c) Observes and maintains ethical standards in dealing with the different stakeholders d) Integrates Christian perspectives and values in all undertakings e) Manifests humility and respect in relating with other people
2. Patriotic	a) Manifests a deep sense of nationalism by integrating history, arts, and culture in their daily lives b) Participates responsibly and collaboratively in the discussion and resolution of issues within local, national, and international contexts c) Engages actively in political, social, economic, and cultural transformation for nation building d) Brings pride and honor to the community and the country e) Patronizes locally produced products and promotes them globally
3. Socially Responsive	a) Demonstrates concern and compassion for the plight of the vulnerable and marginalized sectors of society by participating meaningfully in the process of social transformation b) Engages in ecological advocacies as responsible stewards of God's creation c) Continuously works in solidarity with people and institutions to effect liberating action d) Brings forth awareness and deeper understanding of social realities e) Manifests spirit of volunteerism through social actions that improve communities
4. Globally Competitive	a) Takes progressive responsibility for own learning and development b) Promotes critical and creative thinking, self-knowledge and self-mastery c) Is committed to rise above mediocrity d) Shows dedication, commitment and love for work to attain exceptional results e) Exhibits innovativeness and creativity in various contexts

[1] *Approved in July 2017 during a multi-sectoral committee specifically convened for the purpose of coming up with descriptions for the graduate attributes.*